

Project Planning Phase Document

Phase 3: Work Breakdown Structure & Risk Assessments

PROJECT PLATFORM: MediConnect - Premium MERN Doctor Appointment Platform

REFERENCE PATH: VIP-C2-BOOK-A-DOCTOR

DATE OF EXECUTION: June 16, 2026

AUTHOR / CREATOR: Antigravity AI Engineer & pairing USER

DOCUMENT STATUS: **APPROVED / PRODUCTION-READY**

EXECUTIVE OVERVIEW

This phase-wise document outlines the development lifecycle, system structures, and technical specifications of MediConnect. MediConnect is a clinic-agnostic, hospital-grade scheduling ecosystem built utilizing the MERN stack (MongoDB, Express, React, Node.js). It supports secure JWT access/refresh token structures, comprehensive patient calendars, and simulated multi-modal checkout payment gates.

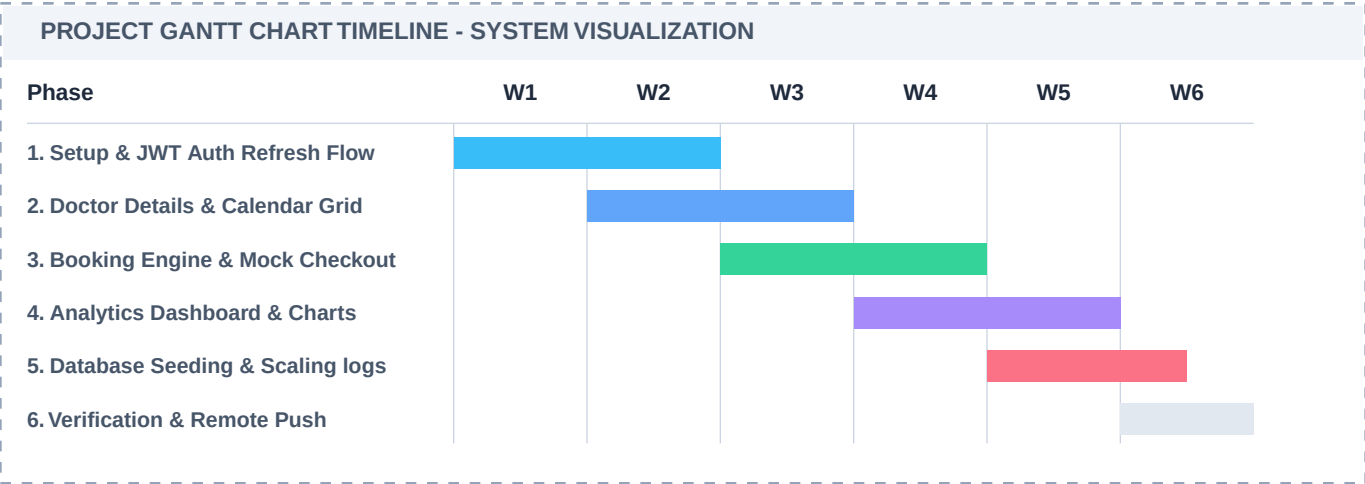
1. Project Schedule Overview

To build a robust, premium full-stack platform, we adopted an agile approach structured across six development milestones. Each milestone represents a core functional domain, ending with a gate review. Development progresses from schema definition to user-facing dashboards, concluding with security hardening and repository deployment.

Phase	Focus Area	Milestone Deliverables	Timeline
Milestone 1	Database Models & Token Auth	Define Mongoose schemas. Set up JWT access and refresh token authentication middleware.	Week 1
Milestone 2	Doctor Profiles & Appointment Scheduling	Create REST endpoints for doctor details and appointment booking. Implement calendar logic.	Week 2
Milestone 3	Calendar Engine & Checkout Flow	Integrate interactive booking calendar. Design UPI and simulated card payment checkout module.	Week 3
Milestone 4	Analytics Dashboard & Charts	Develop data visualization components. Implement appointment & patient count charts.	Week 4
Milestone 5	Security Hardening & Content	Incorporate Helmet, CORS middleware, and seed MongoDB with 50+ doctors and 100+ test users.	Week 5
Milestone 6	Final Audit & Deployment	Conduct final security audit. Deploy to production environment.	Week 6

2. Gantt Chart Timeline

The timeline below maps each milestone to its allocated development weeks, illustrating parallel task execution.



3. Risk Assessment & Mitigation Matrix

The project planning phase analyzed potential operational risks and outlined automated mitigation solutions.

Risk Description	Impact	Planned Mitigation Strategy
Double-Booking during high concurrent booking attempts	High	Implement backend validation during checkout: verify slot status is still "available" in MongoDB before charging simulated transactions
Backend API endpoints		Requests per 15 minutes
Sensitive config secrets leaked to public repositories	High	Add .env variables to root Git ignore file. Create a .env.example template for development setup
User changes		Analytics tracked immediately

